

Seong-Hoon Wang(왕성훈)

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Research Interests: Model Compression, Efficient AI Systems

PROFESSIONAL SUMMARY

Undergraduate researcher focused on Edge AI, efficient deep learning models, and model optimization for embedded systems. Worked on inference pipeline optimization, selective post quantization, and resource-efficient AI systems for embedded environments. Interested in model compression and efficient deep learning for edge devices.

EDUCATION

Soonchunhyang University, Asan, South Korea

B.S. Candidate in Computer Software Engineering

Expected Graduation: Feb. 2028

- Relevant Coursework

Year	Semester	Course Title	Grade
2025	2	Machine Learning	A+
2025	2	Algorithms	A+
2025	2	Information Theory	A0
2025	1	Data Structures	A0
2025	1	Discrete Mathematics	A+
2025	1	Computer Architecture	A+
2024	2	Probability & Statistics	A+
2024	1	Linear Algebra	A+

- Overall GPA: 4.29 / 4.5

RESEARCH EXPERIENCE

INSLab (Intelligence Networking & System Laboratory)

Undergraduate Research Assistant

Soonchunhyang University
2024.04 – Present

- Successfully deployed a YOLOv8n Pose model on the Hailo8 NPU environment and optimized the inference pipeline for embedded systems, reducing NPU utilization by approximately 50% compared to the previous model.
- Designed and applied a learnable preprocessing layer to a MobileNetV3-based Presentation Attack Detection (PAD) model, improving robustness against spoofing attacks; related research is currently in progress.
- Developed a real-time Presentation Attack Detection pipeline for intelligent trash bin users using MobileNetV3 and TAPPAS framework (Gstreamer-based), including model inference and streaming pipeline integration.
- Evaluated selective Post Quantization (PTQ) strategies on the Hailo8 NPU platform, reducing computational resource usage while maintaining comparable detection performance.
- Proposed a YOLO Pose-based architecture for illegal dumping detection for intelligent trash bins and validated its feasibility through deployment on the Hailo8 NPU environment.

MANUSCRIPTS IN PREPARATION

- Manuscript in preparation on Post Quantization (PTQ) strategies for Hailo8-based computer vision models.
- Manuscript in preparation on learnable preprocessing layers for robust Presentation Attack Detection (PAD) in Hailo8-based edge AI environment.

PUBLICATIONS

- Seong-Hoon Wang, SoYeon Lee, Jae-Won Jang and Dae-Young Kim. “특징 중요도 기반 선택적 특징 추출을 통한 경량 위조 매체 판별 시스템” , 대한전자공학회 학술대회논문집, 2026. (Accepted)

AWARDS & HONORS

- Academic Excellence Award (Dean’s List), Soonchunhyang University – Spring 2024, Fall 2024, Fall 2025
- Best Academic Award (Dean's List), Soonchunhyang University – Spring 2025

SCHOLARSHIPS

- Academic Excellence Scholarship, Soonchunhyang University – Spring 2024, Fall 2024, Spring 2025
- Academic Excellence Scholarship (Honorable Mention), Soonchunhyang University – Fall 2025

KEY SKILLS

Programming	Python, C
Frameworks & Libraries	PyTorch
Edge AI	Hailo8 NPU, Gstreamer
Computer Vision	Object Detection, Pose Estimation, PAD
Research Topics	Model Optimization, Quantization, Compression
Tools	Linux, Git
Languages	Korean (Native), English